

name: Handle Missing Object by Using a Substitute

description: A fallback strategy for when a required object cannot be found, which involves using a suitable substitute to complete the task.

skill that demonstrate meta strategies of failure recovery

Workflow

relate to existing skills, showcasing composition capabilities

1. **Exhaustive Search:** First, perform a thorough search for the primary target object. Use a systematic method, such as the Find Object by Prioritizing Likely Locations skill, to check all probable locations.
2. **Confirm Unavailability:** If the object is not found after an exhaustive search, conclude that it is not available in the environment.
3. **Identify a Substitute:** Scan the list of observed objects for a suitable substitute. A good substitute typically belongs to the same general category or can serve a similar function.
4. **Proceed with Substitute:** Continue the task using the identified substitute object as if it were the original target.

Heuristics for Choosing a Substitute

- **Dishware:** If a Plate is missing, use a Bowl.
- **Utensils:** If a Fork is missing, use a Spoon or ButterKnife.
- **Cookware:** If a Pot is missing, use a Pan.
- **Cleaning:** If a Towel is missing, use a Cloth.

When to Use

- Use this skill only after a comprehensive search for the primary object has failed. Do not use it as a first option.

(a) Skill curated by SkillOS for agentic tasks

name: Relate Incenter-Vertex Distance to Triangle Properties

description: Calculate triangle properties like angles or side lengths using the distance from a vertex to the incenter, the inradius, and the semiperimeter.

Workflow

When you know or can calculate the distance from a vertex (e.g., A) to the incenter (I), you can use the following two formulas to find other triangle properties.

Method 1: Find the Angle at the Vertex

Skill that shows branching-out reasoning behavior

1. **Formula:** The distance IA is related to the inradius r and the angle A by:

$$IA = r / \sin(A/2)$$

2. **Application:** If IA and r are known, you can solve for $\sin(A/2)$. From $\sin(A/2)$, you can find $\cos(A/2)$ (using $\sin^2\theta + \cos^2\theta = 1$) and subsequently $\sin(A)$ and $\cos(A)$ using double-angle formulas.

Method 2: Relate to Side Lengths

1. **Formula:** The distance IA is related to the inradius r , the semiperimeter s , and the side a (opposite to vertex A) by:

$$IA^2 = r^2 + (s - a)^2$$

2. **Application:** If IA and r are known, you can solve for $(s - a)$. This provides a direct equation linking the semiperimeter and a side length.

Prerequisite Constraints

- The inradius r must be known.
- The distance from a vertex to the incenter (e.g., IA) must be known or calculable.

(b) Skill curated by SkillOS for reasoning tasks